Module 127-139.

Q NO:- 01.

Compute the factor group $(Z_4 \times Z_6)/\langle(0,1)\rangle$.

Solution:-

Here $\langle(0,1)\rangle$ is the cyclic subgroup H of $Z_4 \times Z_6$ generated by (0,1). Thus:-

$H = \{(0,0), (0,1), (0,2), (0,3), (0,4), (0,5)\}$. Since $Z_4 \times Z_6$ has 24 elements, all the cosets of H must have 6 elements and $(Z_4 \times Z_6)/H$. Remember that, we compute in a factor group by the representative from the original group.

In additive notation, the cosets are:

$H = (0,0)+H, \ (1,0)+H, \ (2,0)+H, \ (3,0)+H.$

Since we can compute by choosing the representatives

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2.

$(0,0)$, $(1,0)$, $(2,0)$ and $(3,0)$ it is clear that $(Z_4 \times Z_6)/H$ is isomorphic. Note that this is what we would expect, since in a factor group modulo $H$, everything in $H$ becomes the identity element; that is, we are essentially setting everything in $H$ equal to zero. Thus, the whole second factor $Z_6$ of $Z_4 \times Z_6$ is collapsed, just leaving the first factor $Z_4$.

Q NO:- 02.

3.

Show that the group $G = Z/12Z$ is not a simple group.

Solution:-

The set $H = \{0, 4, 8\}$ of congruence classes of 0, 4 and 8 modulo 12 is a subgroup of order 3, and it is a normal subgroup since any subgroup of an abelian group is normal. Hence, $G = Z/12Z$ is not simple.

Q NO:-3*

Define commutator subgroup.

Solution:-

Let G be a group, then the set of all the commutators $aba^{-1}b^{-1}$ for all $a, b \in G$ generates a subgroup C of G, that subgroup C is known as the commutator subgroup.

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Q NO:- 04.

S.

Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for all $a, b \in G$, generates a subgroup C of G.

Solution:-

Let $a, b \in G$. Then:

$$(aba^{-1}b^{-1})(aba^{-1}b^{-1})^{-1} = aba^{-1}b^{-1}bab^{-1}a^{-1}$$

$$= e \in C.$$

Since $e = eee^{-1}e^{-1}$ is a commutator.